

JEE MAIN REVISION MASTER: CHEMICAL EQUILIBRIUM 2026 2 QUESTIONS 3.33% WEIGHTAGE ↓33.4%

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Chemical Equilibrium - Core Application	11 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Chemical Equilibrium - Core Application (High Yield — 11 PYQs)

Core Principle: Key problem pattern in Kcet Chemical Equilibrium.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "For the following gaseous reversible reaction: $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_{(3)}\text{B}_{(g)}$ $\Delta_r H = -q$..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "For the reversible reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ When the partial pressure is measured i..." [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The equilibrium constant at 298 K for the reaction $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$ is 100. If the initial concentrations of all the four speci..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

For the following gaseous reversible reaction: $3A_{(g)} + B_{(g)} \rightleftharpoons A_3B_{(g)}$ ($\Delta_r H = -q \text{ kJ}$). The amount of product $A_3B_{(g)}$ is affected by ____

- (A) Temperature alone
- (B) Pressure alone
- (C) Both temperature and pressure
- (D) Temperature, pressure and catalyst

Correct Answer: None

Solution Analysis: For the reaction $3A_{(g)} + B_{(g)} \rightleftharpoons A_3B_{(g)}$ the forward reaction is exothermic because $\Delta_r H = -q \text{ kJ}$. So, heat is released in the forward direction. If temperature is increased, the equilibrium shifts in the backward direction to absorb heat. So the amount of A_3B decreases. If temperature is decreased, the equilibrium shifts in the forward direction. So the amount of A_3B increases. Now look at pressure. On the reactant side, the number of gaseous moles is $3+1=4$. On the product side, the number